Exercise 2 :

a) for $t = 1976$, using cos and sin property yields

$$\hat{\mu}_{1976+\frac{3}{12}} = \hat{\mu}_{t+\frac{1}{4}} = 46,2660 + (-26,7079)\underbrace{\cos\left(\frac{2\pi \cdot 1}{4}\right)}_{=\cos(\pi/2)=0} + (-2,1697)\underbrace{\sin\left(2\pi \times \frac{1}{4}\right)}_{=\sin(\pi/2)=1}$$

$$\boxed{\hat{\mu}_{1976+\frac{1}{4}} = 44,0963}$$

b) A 95 % intervall for $\mu_{1976+\frac{1}{4}}$ is : $44,0963 \pm 1,96 \times 3,719$

$$\Rightarrow \quad \boxed{CI = 44,0963 \pm 7,289}$$

c) Similar to (a) :

$$\hat{\mu}_{1977+\frac{1}{4}} = 44,2660 + (-26,7079)\underbrace{\cos\left(\frac{\pi}{2} + 2\pi(1977)\right)}_{0} + (-2,1697)\underbrace{\sin\left(\frac{\pi}{2} + 2\pi(1977)\right)}_{1}$$

$$\boxed{\hat{\mu}_{1977+\frac{1}{4}} = 44,0963}$$

$$\hat{\mu}_{2009+\frac{1}{4}} = 44,2660 + (-26,7079)\underbrace{\cos\left(\frac{\pi}{2} + 2\pi(2009)\right)}_{0} + (-2,1697)\underbrace{\sin\left(\frac{\pi}{2} + 2\pi(2009)\right)}_{1}$$

$$\hat{\mu}_{2009+\frac{1}{4}} = 44,0963$$

We have exactly the same result. The forcast depends only on the month regardless of the year.